3 The Iron Age

Topics covered

- The 'Ages' of human history
- The changes of the Iron Age
- Usefulness of iron
- Civilizations of the Iron Age

These are some materials that the earliest human beings used. Talk about which of these we use even today and how we use them.

• stone • wood • bone • horn • copper • bronze • iron

THE 'AGES' OF HUMAN HISTORY

The materials used by early human beings are still useful for us today. As you know, we have learnt about the life of these human beings through the discovery of artefacts used by them. To make it easier to study early human history, historians have divided it into 'ages'. An age is a very long period of time. This table tells you some things about the ages. The blank section of the table (on the next page) is what you will learn about in this chapter.

Age	Materials used	Life of the people
Stone Age	• Stone, wood, bone, and horn • In the later part, people learnt to make pottery using clay.	• Hunting and gathering • Later, farming developed along with animal rearing and trade. • Villages and towns grew.

Age	Materials used	Life of the people
Copper and Bronze Ages	All the materials used earlier, and also: • Copper • Bronze made by mixing copper with tin	• Various **civilizations** began to develop. They had good agriculture and the earliest cities. • Cities and civilizations traded amongst themselves to get the goods that they needed.
Iron Age	The materials used earlier, and also: • Iron	

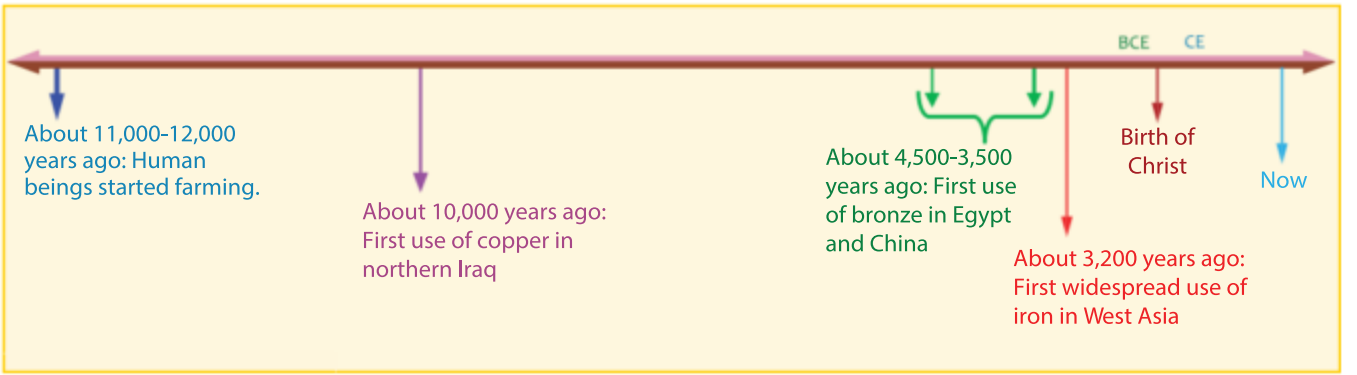
Though early history has been divided in this way, one age did not come to an end neatly before another one started. When people started using bronze, they were also still using stone and copper. Similarly, when people started using iron, they were also still using bronze. The **Iron Age** was the time in early human history when people began to use tools and weapons made of iron.

These different ages did not happen at the same time all over the world. For example, people in Egypt were making bronze articles 4,500 years ago, whereas in China this happened 3,500 years ago. Similarly, the Iron Age happened at different times in different parts of the world.

Did you Know?

Very early iron artefacts—beads and a dagger—have been found in Egypt. Analysis has shown that these were made from iron found in meteorites (stones that fall to earth from outer space). Ancient Egyptians called this iron *ba-en-pet* which means 'Stone or Metal from Heaven'.

Historians think that the earliest widespread use of iron started about 3,200 years ago or around 1200 BCE in West Asia, in the area that is now Turkey (see your atlas). For a long time, only these people knew how to use iron, but this knowledge gradually spread to other parts of the world. The timeline shows you some important events in human history.



USEFULNESS OF IRON

Iron was an extremely useful metal for human beings. You will understand this if you look at the differences between bronze and iron.

A lump of iron ore—the raw state of iron

- Iron is a much harder metal than copper or bronze so it was better for making tools and weapons.
- Iron is found in many parts of the world, whereas tin, which is used to make bronze, is not as widespread.
- All metals melt when they are heated to a high temperature. Bronze melts at a much lower temperature than iron. Early people were able to melt bronze and pour it into moulds to make weapons or tools. However they could not heat iron enough to make it melt. So it was heated and when it was soft, it was beaten into the tools or weapons that were required. This process made their tools and weapons very strong.

Melted bronze being poured into moulds

Discuss

Why were stronger and harder tools better for early human beings?

Several iron artefacts from early human history have been found. These show that people had learnt to make useful objects, particularly tools and weapons, from this metal. Some of the artefacts that have been found are axes, ploughshares (used to dig fields), sickles (used to harvest crops), nails, swords, spears, arrowheads and pots.

Wooden handles were fitted to these (left) iron sickles and (right) iron axes.

THE CHANGES OF THE IRON AGE

Agriculture

People used iron axes to clear forests so that they had more land for farming. Iron ploughshares, fixed to wooden ploughs, could dig deeper and turn the soil better than earlier tools. This helped to grow better crops, so more food could be produced. Iron sickles were better for harvesting (cutting) the ripe crops.

Buildings and crafts

Iron helped to improve crafts as well as building activities. Some artefacts that have been found are iron cooking pots and nails.

Trade

As tools improved, more food and other goods were produced than were actually needed. This is called a **surplus** (extra). Towns and villages sold their surplus to each other. This is called trade. Trade between different regions increased. With increasing trade, **money** began to be used. Some early coins were of iron, but later coins were of metals like silver and gold. Coins were also made of a mixture of silver and gold.

Early coins of iron, copper, silver and gold

Cities and urban centres

With improved agriculture, everyone did not need to be a farmer. A few people could produce the food that was needed. Many people were free to do other things. Cities and **urban** centres (areas connected to cities) grew. Some of these were very grand, with magnificent buildings, art, craft and literature. They became the centres for trade as well.

Weapons and warfare

Iron was extremely useful in making better weapons like swords, spears and arrowheads. Wooden wheels on war chariots were fitted with iron **spokes** and **rims**. This made the chariots faster. People with iron weapons could win battles easily against people who did not know about iron or its uses. Thus iron weapons helped people to gain more **territory** (area) over which they could rule. Kingdoms become bigger.

Iron arrowheads like these were fitted to wooden arrows.

Think and answer

The Hittites in West Asia were the first to use iron extensively, but they tried to prevent other people from learning about it. Why do you think they did this? (Hint: weapons)

CIVILIZATIONS OF THE IRON AGE

With the growth of cities and urban centres, civilizations developed. '**Civilization**' means that a place is highly developed in areas like art, trade, writing, architecture, etc. All civilizations of the ancient world did not exist at the same time. Many civilizations developed before the Iron Age. Two civilizations of the Iron Age were the Greek and the Roman civilizations. You will learn more about them in later classes. The maps show you their locations—both were in areas around the Mediterranean Sea.

Ancient Greece, at the time of the Greek civilization, had many independent city states spread over the area of this map. Some cities are marked in the map.

The huge Roman empire in 117 CE —Rome was the chief city of the empire. Find the Mediterranean Sea in both maps.

The discovery and use of iron was an extremely important development in human history. Iron is still extensively used today.

Did you Know?

The best steel in the Roman civilization was Seric steel from southern India. Trade in this metal was part of Indian trade with Rome.

Values and Life Skills

- Think of all the things that you use or see around you. Identify (with the help of your teacher) the ones that are made of iron.
- Discuss the importance of the discovery of iron in our lives. This will help you to understand how the past is with us even today.

REMEMBER

1. Early human history is divided into long periods of time called 'Ages' based on the materials that people used.
2. The various Ages happened at different times in different places.
3. The Iron Age is the period when people began to use iron to make tools and weapons.
4. Iron is a hard metal and so it was used to make strong weapons and tools.
5. The use of iron changed human lives in many ways.
6. Two Iron Age civilizations were the Greek and Roman civilizations.

NEW WORDS

Civilization	a state of human society that is very developed and organized
Meteorite	a piece of rock from outer space that hits the earth's surface
Surplus	an amount that is extra or more than you need
Urban	connected with a town or city
Spoke	a thin straight piece of metal that connects the centre of a wheel to its outer edge, for example on a bicycle wheel
Rim	the edge of something in the shape of a circle, for example, the rim of a cycle tyre
Territory	land that is under the control of a particular country or ruler

EXERCISES

A. Fill in the blanks.

1. The ..... of human history lasted for thousands of years.
2. Cities and civilizations ..... amongst themselves to get the goods they needed.
3. When people started using iron, they continued to use .....
4. With improved agriculture, people could produce more ..... than was required.
5. ..... means that a place is highly developed in areas like art, trade, writing, etc.
6. Two Iron Age civilizations are the ..... and ..... civilizations.

B. Match the following.

A	B
1. territory	a. connected with a town or city
2. surplus goods	b. area that is under the control of a particular ruler
3. iron spokes and rims	c. mixture of copper and tin
4. bronze	d. trade increased
5. urban	e. made war chariots faster

C. Write true or false.

Correct the false statements in your book.

1. The artefacts used by early people tell us about them and their lives.
2. When one Age ended, another one started.
3. The different Ages started at the same time all over the world.
4. Iron ore is more widely found than tin.
5. During the Iron Age many people had time to do things other than farming.
6. Trading centres were located far away from the cities.

D. Choose the correct answer.

1. Farming and animal rearing developed during the later part of the Bronze Age/ Stone Age.
2. Bronze/Iron is better for making tools and weapons because it is a harder metal.
3. The knowledge of how to use iron spread very quickly/was known only to some people for a long time.

4. Cities of the Iron Age were quite small/had grand buildings, art and craft.
5. All the ancient civilizations existed/did not exist at the same time.

E. Answer the following questions.

1. Why is early human history divided into 'Ages'?
2. What is the Iron Age?
3. When and where did the Iron Age first appear?
4. Explain how each of these iron tools helped to improve agriculture: axes, ploughshares, sickles.
5. How was iron made into tools and weapons?
6. Arrange these phrases in the right order to show how trade developed in the Iron Age: Better tools; Coins of iron, silver and gold began to be used; Trade increased; More goods produced.
7. What were the effects of better weapons made of iron?

THINGS TO DO

A. Picture study

This is a stone carving from Turkey of a battle scene showing two warriors in a chariot.
Talk about what else you can see in this picture.

SI **B. On your own**

Metals of the metal age
Bring objects made of copper, bronze and iron to class. You can also look at silver and gold objects. Talk about their differences that you notice.

TP **C. In a group**

Work in small groups to put together a PowerPoint or chart presentation of:

- the sources from the Stone and Iron ages
- how these sources show the change in people's lives

WEBLINK

- The Iron Age: https://www.theschoolrun.com/homework-help/the-iron-age (Accessed on 20 July 2017)